

Class 9: Candy Mini-Project

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Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA).

Data Import

The data is all related to Halloween candy and is from the 538 website.

```
candy_file <- "candy-data.txt"
candy <- read.csv(candy_file, row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

What is your favorite candy?

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy is 100 grand.

```
candy["100 Grand", ]$winpercent
```

```
[1] 66.97173
```

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Side-note: the `skimr::skim()` function

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, winpercent is on a different scale because it ranges roughly from 20–85, while most columns are 0/1.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

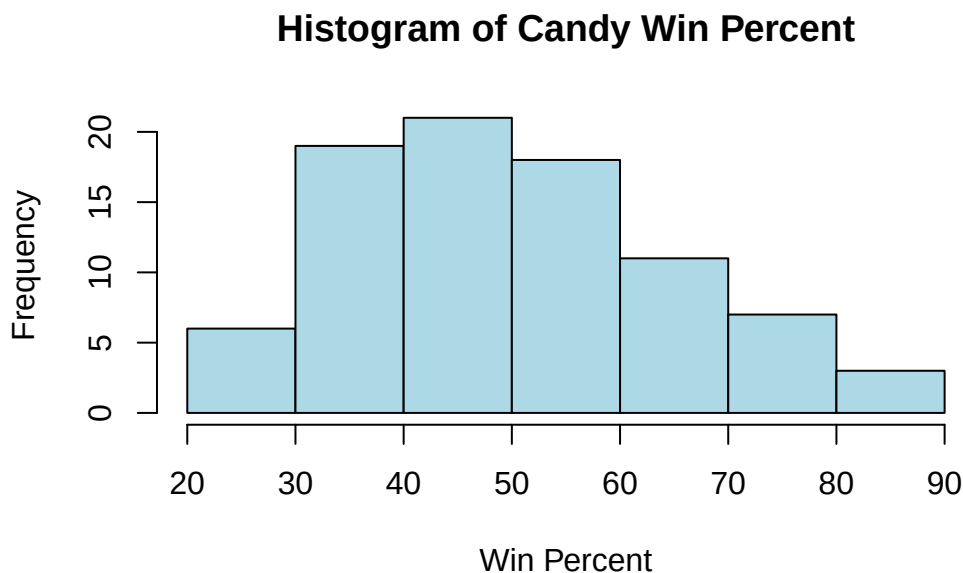
In chocolate, 1 means the candy contains chocolate and 0 means it does not.

Exploratory analysis

A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function `hist()`. Alternatively, you can use `ggplot()` with `geom_hist()`. Either works well in this case and (as always) it's your choice.

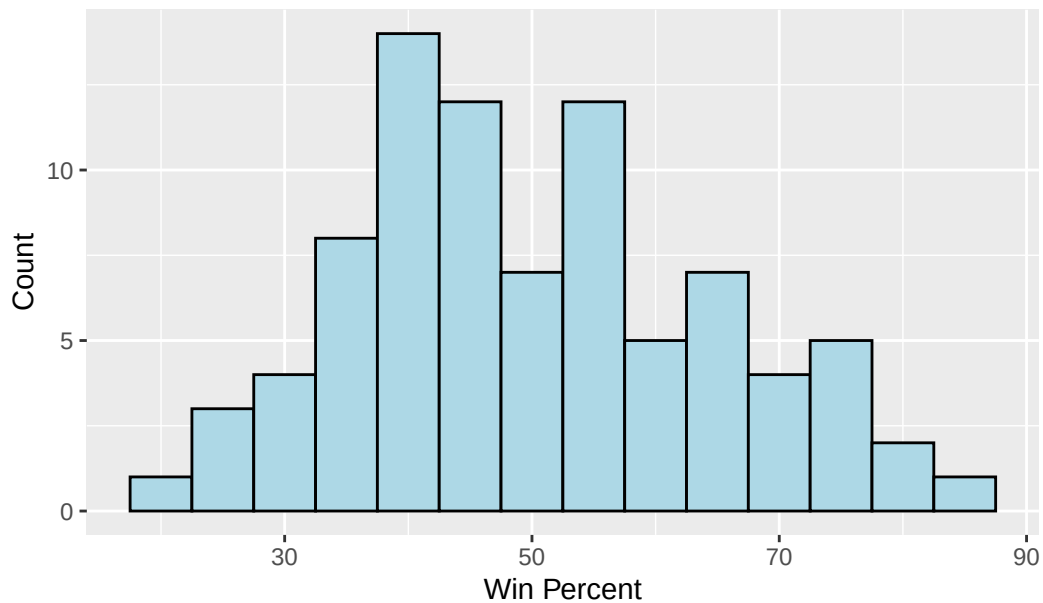
Q8. Plot a histogram of `winpercent` values using both base R and `ggplot2`.

```
hist(candy$winpercent,  
     main = "Histogram of Candy Win Percent",  
     xlab = "Win Percent",  
     col = "lightblue")
```



```
library(ggplot2)  
  
ggplot(candy) +  
  aes(winpercent) +  
  geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +  
  labs(title = "Histogram of Candy Win Percent",  
       x = "Win Percent",  
       y = "Count")
```

Histogram of Candy Win Percent



Q9. Is the distribution of winpercent values symmetrical?

```
# Summary stats to judge center  
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

```
# Mean and median (helps determine symmetry)  
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
median(candy$winpercent)
```

```
[1] 47.82975
```

No, it is slightly skewed to the right. (Since mean > median).

Q10. Is the center of the distribution above or below 50%?

Since the mean = 50.32, the center of the distribution is slightly above 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# Mean winpercent for chocolate candies
mean_chocolate <- mean(candy$winpercent[as.logical(candy$chocolate)])

# Mean winpercent for fruity candies
mean_fruity <- mean(candy$winpercent[as.logical(candy$fruity)])

mean_chocolate
```

```
[1] 60.92153
```

```
mean_fruity
```

```
[1] 44.11974
```

Chocolate candies are ranked higher on average than fruity candies.

Q12. Is this difference statistically significant?

```
t.test(
  candy$winpercent[as.logical(candy$chocolate)],
  candy$winpercent[as.logical(candy$fruity)]
)
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Based on our t-test, which yielded a small p-value of 2.871e-08, the difference is highly significant. So, chocolate candies are significantly more popular than fruity candies.

Overall Candy Rankings

Let's use the base R `order()` function together with `head()` to sort the whole dataset by a variable/column of interest. Or if you have been getting into the tidyverse and the dplyr package you can use the `arrange()` function together with `head()` to do the same thing and answer the following questions:

Q13. What are the five least liked candy types in this set?

```
head(candy[order(candy$winpercent), ], n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$winpercent, decreasing = TRUE), ], n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup				0	0	0		0.720

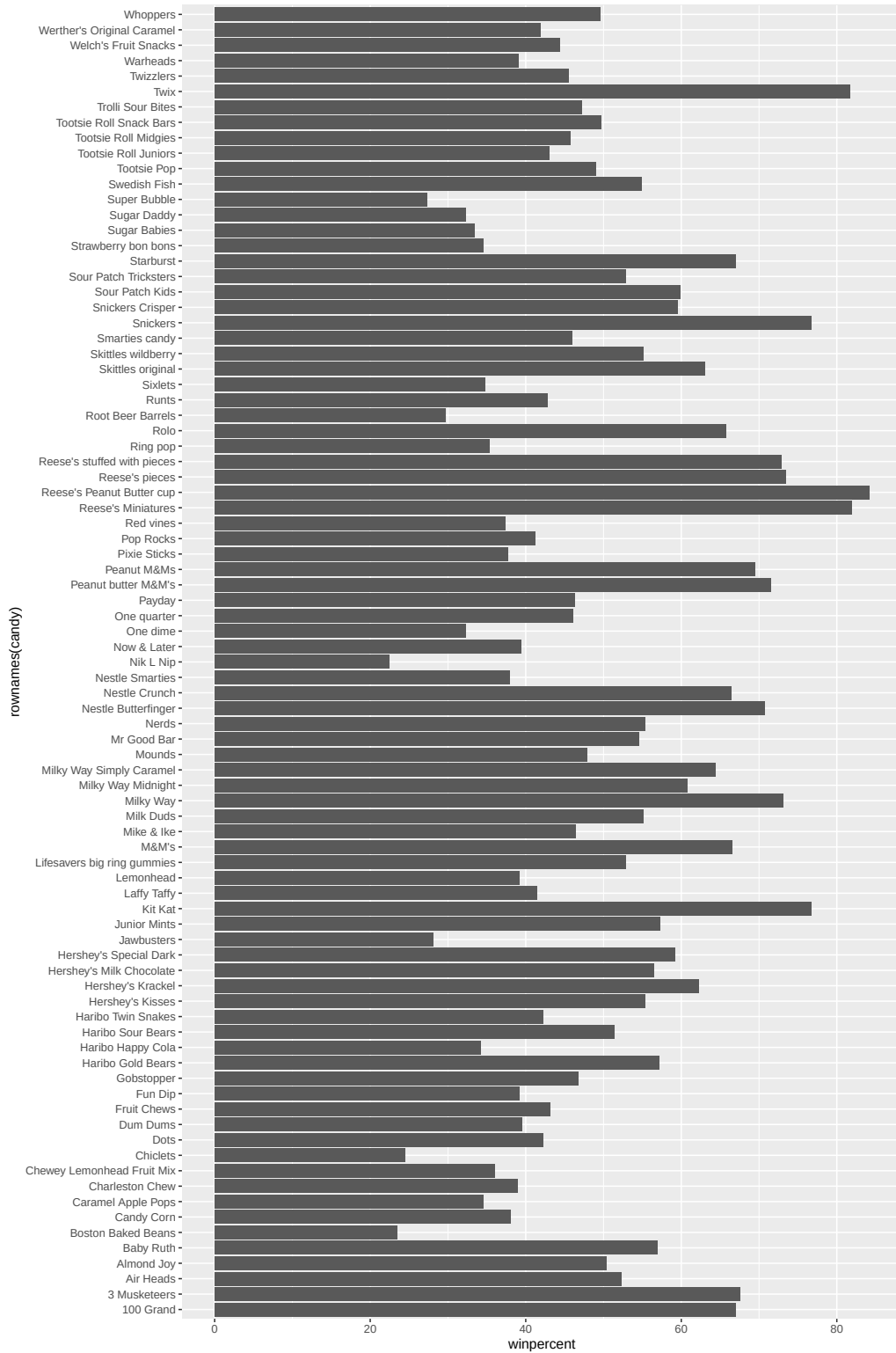
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

Q15. Make a first barplot of candy ranking based on winpercent values.

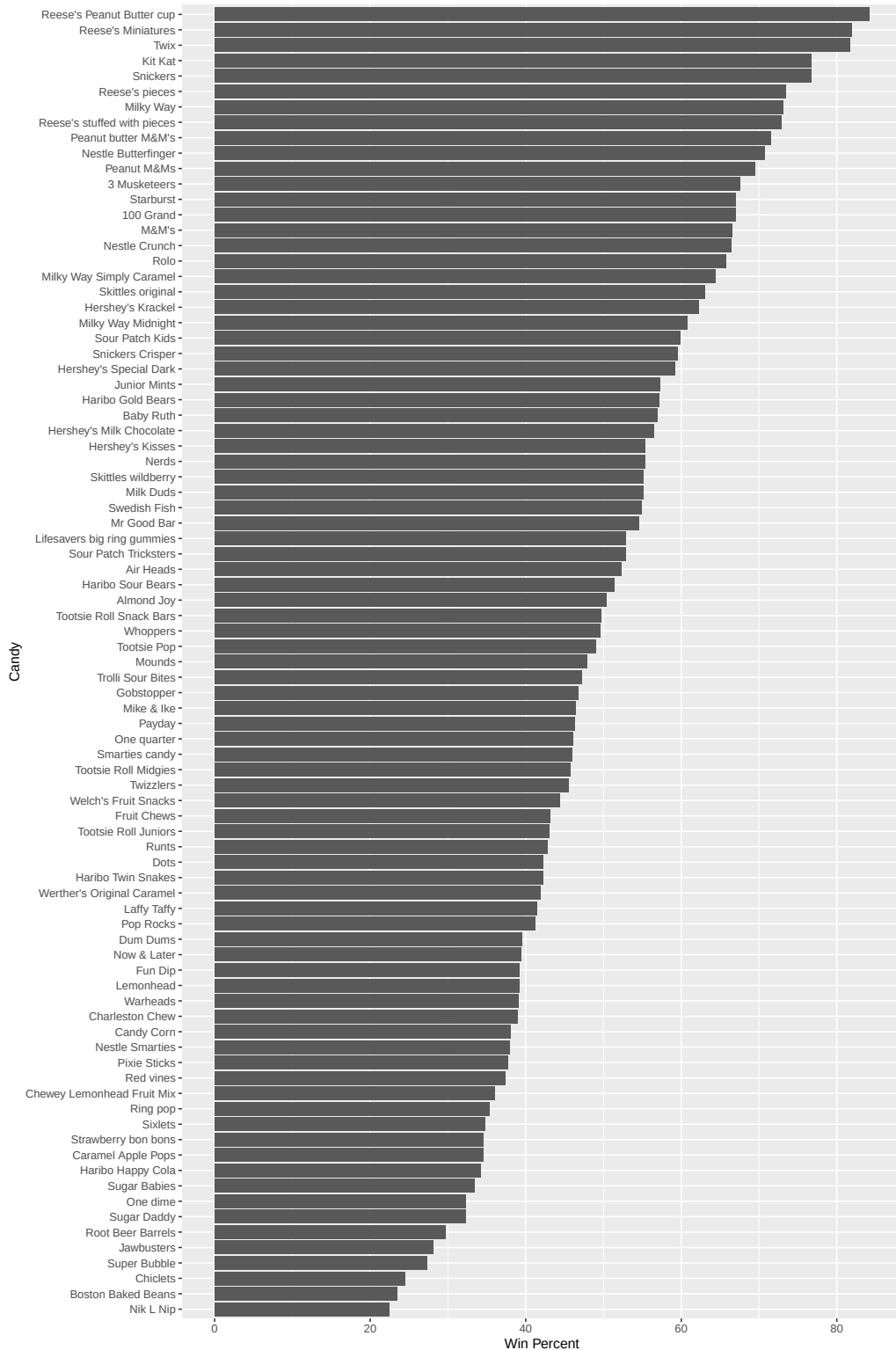
```
library(ggplot2)

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

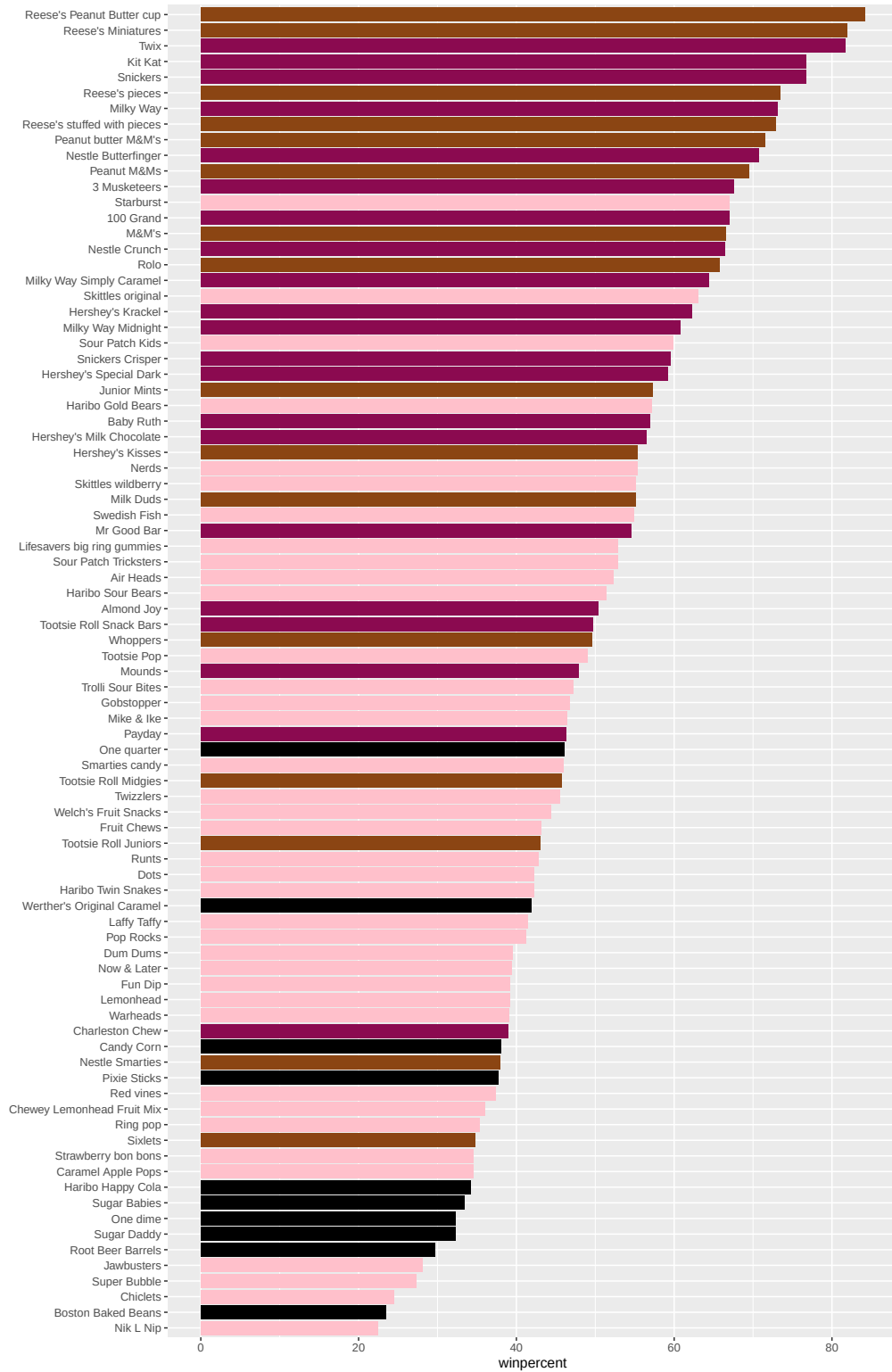
```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col() +  
  ylab("Candy") +  
  xlab("Win Percent")
```



Add some color

```
my_cols = rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate4"
my_cols[as.logical(candy$bar)] = "deeppink4"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill = my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Sixlets.

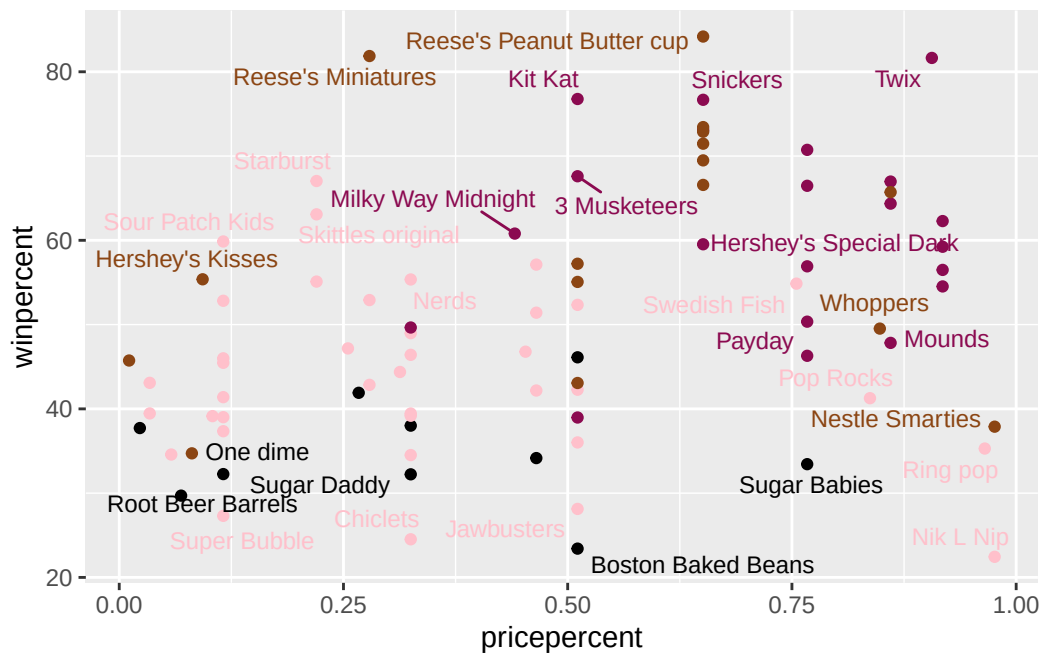
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(pricepercent, winpercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
candy[ord, c("pricepercent", "winpercent")][1:5, ]
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

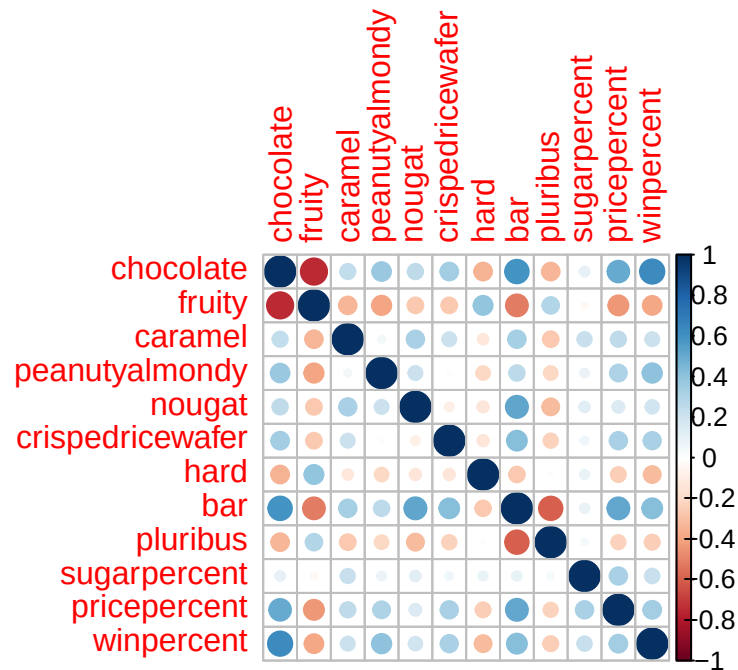
The least popular among these is the Nik L Nip because it has the lowest winpercent.

Exploring the correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```

Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables that are anti-correlated are chocolate and fruity. This means that candies that contain chocolate are very unlikely to be fruity, and vice versa.

Q23. Use your corrpplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

The variables that are expected to have the largest contributions to PC1 are chocolate, bar, winpercent, and pricepercent in the positive direction, and fruity and hard in the negative direction. This suggests that the main variation in the dataset separates chocolate-based candies from fruity candies.

Principal Component Analysis (PCA)

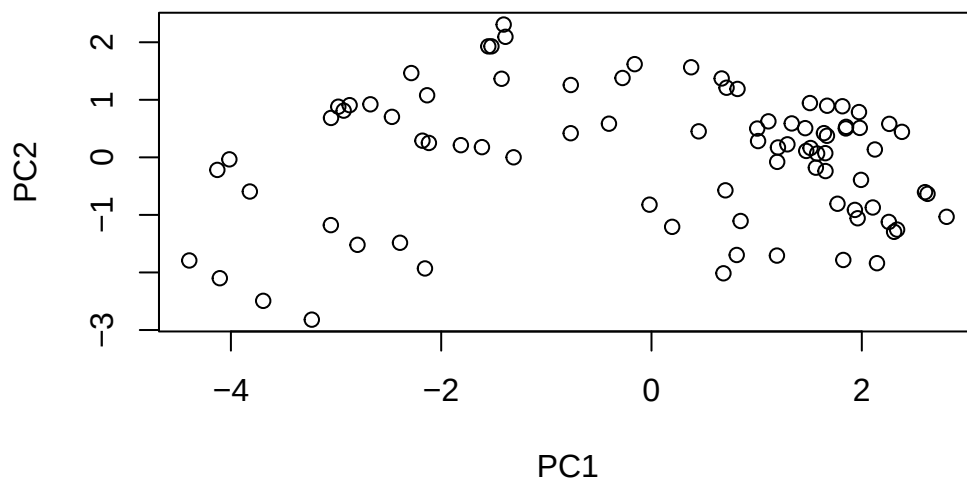
```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

Importance of components:

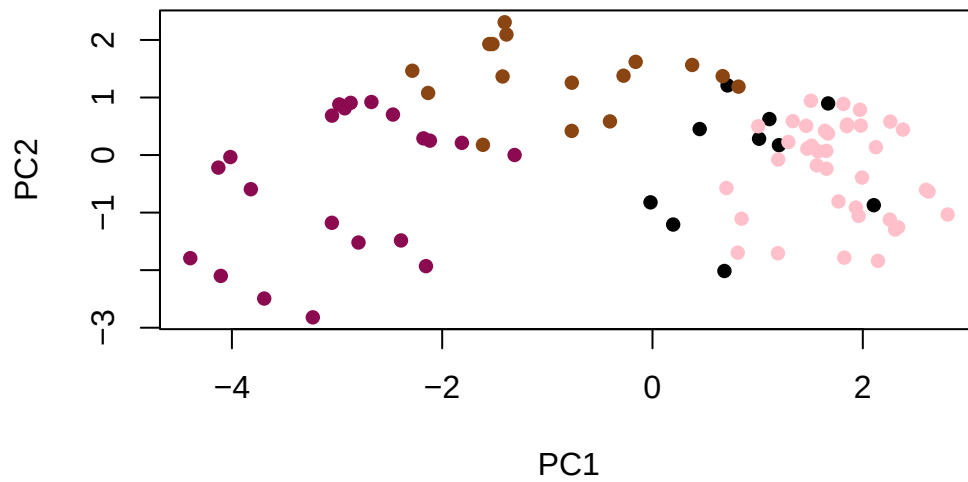
PC1 PC2 PC3 PC4 PC5 PC6 PC7

Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369
	PC8	PC9	PC10	PC11	PC12		
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760		
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317		
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000		

```
plot(pca$x[, 1:2])
```



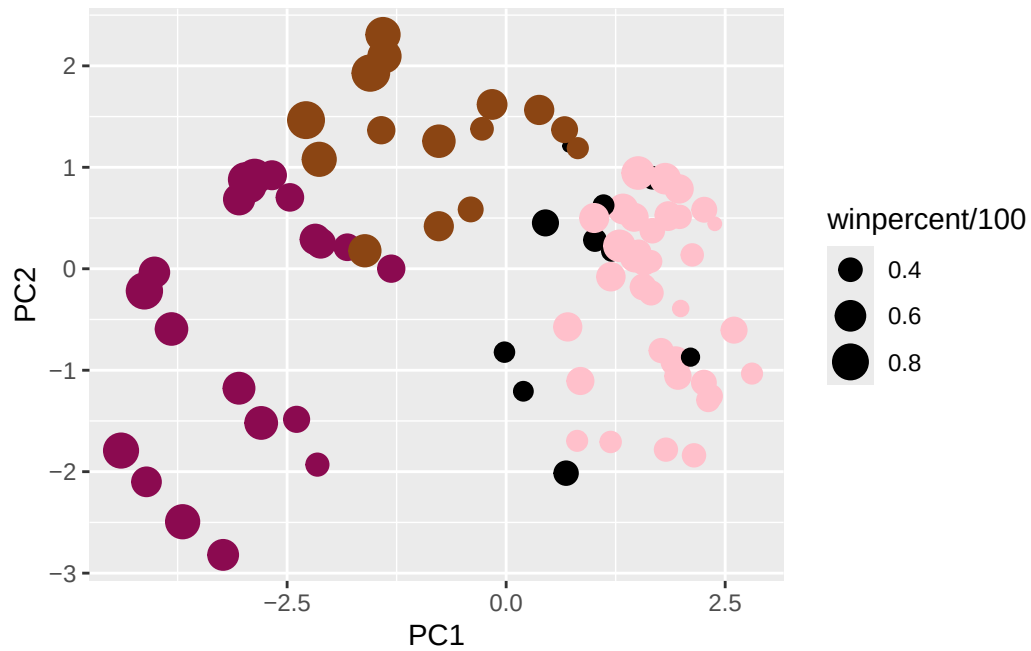
```
plot(pca$x[,1:2], col = my_cols, pch = 16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
```

```
p
```

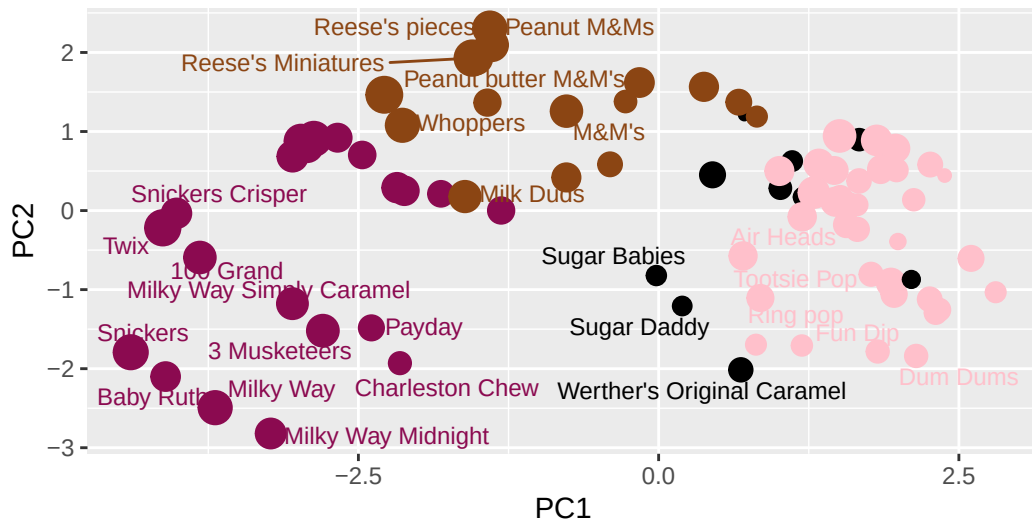


```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),

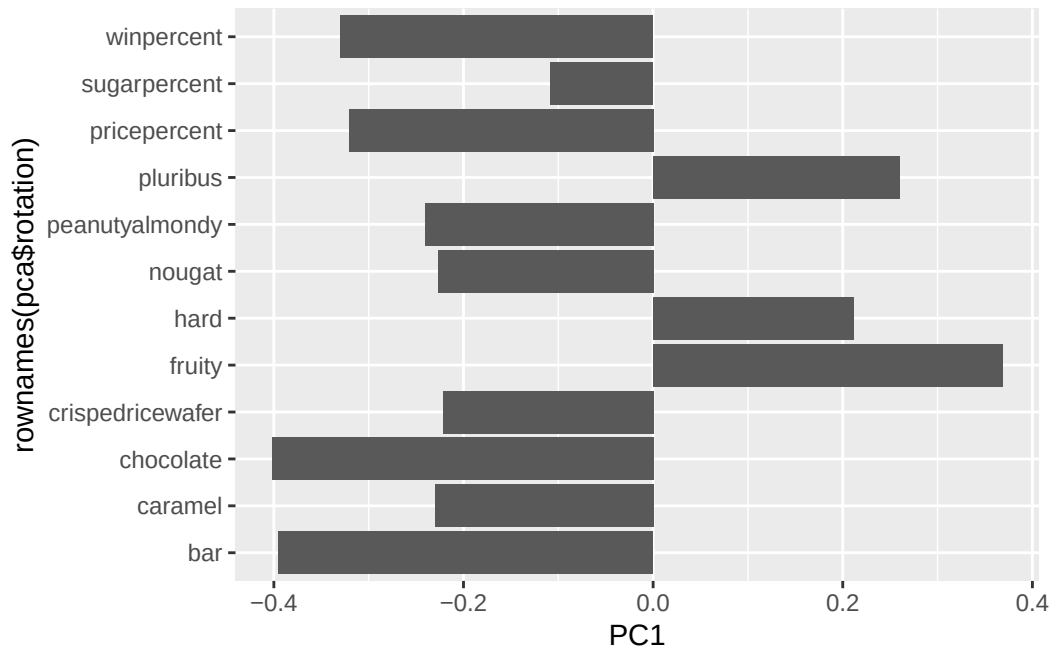


Data from 538

```
#library(plotly)
```

```
#ggplotly(p)
```

```
ggplot(pca$rotation) +  
  aes(PC1, rownames(pca$rotation)) +  
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables that are picked up strongly by PC1 in the positive direction include chocolate, bar, winpercent, and pricepercent. This makes sense because chocolate candies tend to be more popular and often more expensive. This relationship was previously highlighted in the correlation matrix, where chocolate showed a positive correlation with winpercent and pricepercent.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A winning candy is typically chocolate-based, often in the form of a candy bar, and may include ingredients such as peanuts, caramel, or wafers. The visualizations show that top-ranked candies are mostly chocolate, while the correlation analysis demonstrates that chocolate is positively associated with popularity. The PCA further supports this by separating chocolate and fruity candies along the first principal component, confirming that chocolate candies tend to be more popular overall.